# Department of CSE(AIML), PES University
# UE26CS151A -Python for Computational Problem Solving
# Laboratory Week 6

**Note:** Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. The observation record must be signed by the faculty in charge. For every laboratory session, the observation component carries **marks**.

Problem Statements

**Q1. Display Lines Starting with a Given Letter** Write a program to read a text file and display all lines that start with a letter entered by the user.

- **User Input:** File name (string) and starting letter (string) entered by the user.
- **Data Types:** Use string for lines and characters.

**Q2. Find the Student with the Highest Marks Using a Dictionary** Write a program to store student names and marks in a dictionary and display the name and marks of the student who scored the highest marks.

- **User Input:** Student names (string) and marks (int) entered by the user.
- **Data Types:** Use dict with string keys and int values.

**Q3. Calculate the Total Amount for Each Order from a CSV File** Write a program to read product names, quantities, and prices from **week_6_set3_q3.csv** and display the total amount for each order by multiplying its quantity by its price.

- **User Input:** No user input required.
- **Data Types:** Use string for product names, int for quantities, and float for prices.
- **File Reference:** Refer to **week_6_set3_q3.csv** for input.

**Q4. Count the Frequency of Each Word in a Text File** Write a program to read a text file and use a dictionary to display each unique word and its frequency.

- **User Input:** File name entered by the user.
- **Data Types:** Use dict with string keys (words) and int values (frequency).

**Q5. Find Products Within a Given Price Range from a CSV File** Write a program to read product names and prices from **week_6_set3_q5.csv** and display the products whose prices fall within a user-entered minimum and maximum price.

- **User Input:** Minimum price (float) and maximum price (float) entered by the user.
- **Data Types:** Use string for product names, float for prices.
- **File Reference:** Refer to **week_6_set3_q5.csv** for input.

**Q6. Merge Two Dictionaries Containing Student Marks** Write a program to create two dictionaries containing student names and marks, merge them into a single dictionary, and display the result. If a student appears in both dictionaries, retain the marks from the second dictionary.

- **User Input:** Student names (string) and marks (int) entered by the user for both dictionaries.
- **Data Types:** Use dict with string keys and int values.

**Q7. Find the Longest Word in a Text File** Write a program to read a text file and display the longest word present in it. If multiple words have the same maximum length, display the first such word.

- **User Input:** File name entered by the user.
- **Data Types:** Use string for words, int for length comparison.

**Q8. Write a Python program that asks the user to enter the names of three cities and their populations. Store the data in a list of lists and write the results into a CSV file named week_6_set3_q8.csv.**

- **Data Types: string for city names, int for population.**

Prepared By,
**Prof. Adithi S**
CSE Dept RR Campus